⇒ Types of Regression.

LAB

1. Linear Regression:

We have a continuous relation. like if one-axis increases, the other is also increase.

- It has two types ← Univariant, Multivariant

2. Logistic Regression:

Comes under in Classification like We have options Yes, No etc

⇒ If our task is able to use traditional Machine learning models then we don't need to use Deep Learning Model. Only if 100% accuracy is required, For example health.

⇒ Multi-Modality:

- Different forms of data (tabular, audio, video, images) etc combined in one.

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⇒ Univariate Linear Regression:

- Formula: $y = \beta_0 + \beta_1 x$ (β_0 → intercept, β_1 → slop, x → difference, y → predictive answer)
- X = Feature
- Y = Predictive value
- Root Mean square ~~value~~ Error is used to test this model. The value which comes from this test means it is the difference between ... ~~the~~ ac
- R2 Score: Tell how efficiently model understand the data. Its value comes in between 0 and 1.

⇒ Multivariant Regression.

- More numbes of Features, One predictive value.
- But when we do plotting, ~~the~~ we select only one attribute, and the line yes straight because it comes under the linear Regression so we need straight line.
- we don't do plotting foo this.
- ~~finding the cor~~. So we do correlation (which feature create the maximum output value.)

⇒ More nature of problems = More training models.

⇒ If data is noisable or data is missing = Use decision trees.

⇒ For Probability = Naive Byes

⇒ ~~for~~ Linear Data means we separate data with lines easily, SPV is used.

⇒ Non-Linear Data, we do not distributed with the straight line.
- KNN model is used

⇒ Model Generalization = Different environment encounters ~~and~~ not only few, while testing.

Types of

⇒ Dataset :

1) Tabular Data
2) For Images
3) Time Series (Any Data link with time)
   - Models used RNN, LSTM, transformers, auto encoder
4) NLP (For text language)

⇒ ANN → RNN → LSTM → Autoencoder → Attention mechanism
↳ Transformer → LNN } These are Deep Learning models.

⇒ Token = It is a word.

⇒ Machine Learning vs Deep Learning:

Machine Learning	Deep Learning
• Need explicit features	Not need explicit features
• Architecture	"
• Need less data distribution to train	Need more data distribution for train